# SNP CrystalBridge & Transformation Cockpit

## Data Object Creation, Export, Import & Rules Master Guide

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**Target Audience:** SAP Technical Consultants, Migration Leads, ABAP Engineers

### Executive Process Summary

End-to-end data transformation pipeline from ECC Source to S/4HANA Target

This technical document details the complete step-by-step procedure for configuring, extracting, transforming, and importing data using **SNP CrystalBridge (Transformation Cockpit / TC2019)**. It includes exact UI snapshots, architecture flows, reader methods, rule hierarchies, and import modes.

# 01 Architecture & Data Object Creation

System landscape, Shadow DDIC, object header, and table connections

## 1.1 Landscape Foundation & Shadow DDIC

Before creating Data Objects, ensure RFC connections are established across Control, Source (ECC), and Target (S/4HANA) systems, and start the TC Daemon via `/SNP/TE01_DAEMON`. The Shadow DDIC synchronizes table definitions and manages schema deltas (e.g. MATNR 18 to 40 characters or KONV to PRCD_ELEMENTS).

## 1.2 Creating Data Object Header

Data Objects act as logical containers grouping root business tables with all dependent child tables, selections, and local transformation logic.

| Parameter | Setting / Value | Technical Function |
|---|---|---|
| **Object ID** | DO_SD_SALES_ORDER | Unique uppercase identifier in Transformation Cockpit. |
| **Title** | Sales Orders Header & Items | Human-readable description of the migration entity. |
| **Status** | A (Active) | Only active objects are compiled into transformation blocks. |
| **Data Type** | T (Transaction) | Categorizes data into Transaction (T), Master (M), or Customizing (C). |
| **Flags** | Relevant for Export / Import, NZD | Enables export/import inclusion and near-zero downtime logging. |
| **Predecessors** | DO_BC_ADDRESS, DO_SD_CUSTOMER | Enforces sequence (e.g., addresses must exist before customers/orders). |

### Transformation Cockpit Snapshot: Data Object Creation

```
+=============================================================================================================+
| [SNP Transformation Cockpit] - Data Object Maintenance: DO_SD_SALES_ORDER                [Client 100]  |
+=============================================================================================================+
| Object ID        : [ DO_SD_SALES_ORDER          ]  Status: (o) A - Active   ( ) I - Inactive          |
| Title            : [ Sales Orders - Open and Historical Documents                              ]       |
| Data Type        : [ T ] (T: Transaction, M: Master Data, C: Customizing)                              |
| Execution Flags  : [X] Relevant for Export    [X] Relevant for Import    [X] NZD Compliant            |
| Predecessors     : Export: DO_SD_CUSTOMER    | Import: DO_BC_ADDRESS -> DO_SD_CUSTOMER                |
+=============================================================================================================+
```

## 1.3 Table Connections & Reader Selection (Scout vs. Direct Reader)

Child tables are assigned under the root table (e.g., VBAK -> VBAP -> VBEP). SNP provides two specialized extraction readers:

| Reader Type | Mechanism | When to Use | Behavior |
|---|---|---|---|
| **Scout Reader (SR)** | FOR ALL ENTRIES on parent export | Standard parent-child hierarchies | Processes in 2 steps. Dependent table waits for parent export. |
| **Direct Reader (DR)** | INNER JOIN SQL statement | Massive child tables (KONV, BSEG) | Single SQL statement. Fully independent parallel extraction. |

### Transformation Cockpit Snapshot: Table Hierarchy & Readers

```
+=============================================================================================================+
| [-] [DO] DO_SD_SALES_ORDER  [Status: ACTIVE] [Type: T - Transaction]                                        |
|     |-- [T] VBAK  (Sales Header)    [Root Selection Table]   Reader: Scout (SR)    Status: ACTIVE       |
|     |   |-- [T] VBAP  (Sales Item)      [Dependent Table]    Reader: Scout (SR)    Status: ACTIVE       |
|     |   |   |-- [T] VBEP  (Schedule)    [Dependent Table]    Reader: Scout (SR)    Status: ACTIVE       |
|     |   |   \-- [T] KONV  (Pricing)     [Dependent Table]    Reader: Direct (DR)   Status: ACTIVE       |
|     |   \-- [T] VBKD  (Business Data)   [Dependent Table]    Reader: Scout (SR)    Status: ACTIVE       |
+=============================================================================================================+
```

# 02 Selections, Export Rules & Events

Filter criteria, domain rules, and source lifecycle event hooks

## 2.1 Filtering & Selection Strategies

- **Field Selection (Local):** Direct filter applied to table fields (e.g., `VBAK-AUART IN ('OR', 'QT')`).
- **Global Selection (Universal):** Universal filter applied to a Domain (e.g., Domain `BUKRS = 1000, 1100`) inherited across all tables.
- **Free Selection (SQL Expert Mode):** OpenSQL WHERE-clause for complex multi-field criteria.

## 2.2 All Possible Export Rules & Events

Export rules run on the source ECC system during extraction. Events allow custom ABAP logic at defined execution moments:

| Event / Rule | Level / Scope | Parameters | Typical Technical Purpose |
|---|---|---|---|
| **Process Init** | Run start | Memory buffers | Initializes variables and clears temporary export caches. |
| **Before Table** | Table start | IV_TABNAME | Checks table readiness, allocates buffer tables. |
| **Before Package** | Package start | IT_PACKAGE_KEYS | Pre-allocates chunk size, starts performance timer. |
| **Before Record** | Each row | IS_RECORD (Read) | Evaluates record validity across multiple fields. |
| **Global Export Rule** | Domain level | CV_VALUE, IS_RECORD | Address collection (ADRNR), GDPR data scrambling. |
| **After Record** | Each row | CS_RECORD, EV_SKIP | Set EV_SKIP = 'X' to drop obsolete/closed records. |
| **After Package** | Package end | CT_PACKAGE | Performs batch aggregations and checksum verification. |
| **After Table** | Table end | IV_TOTAL_ROWS | Writes extracted counts to execution tracking tables. |

### Transformation Cockpit Snapshot: Export Event Code Editor (EV_SKIP)

```
*---------------------------------------------------------------------*
* SNP Export Event: After Data Record Processing (Table: VBAK)
* Logic: Skip completely closed/cancelled orders older than 2020
*---------------------------------------------------------------------*
IF cs_record-gbstk = 'C' AND cs_record-erdat < '20200101'.
  ev_skip = 'X'.  " Excludes record from being written to export file
  EXIT.
ENDIF.
```

## 2.3 Export Block (EXP) Creation & Execution

The Export Block packages active data objects, allocates parallel work processes (e.g., 12 RFC workers), and divides data into compressed package files (e.g., 50,000 rows/package) stored at `[share]///pkg_*.dat.gz`. Files are immutable and never overwritten.

### Transformation Cockpit Snapshot: Export Execution Monitor

```
+=============================================================================================================+
| [SNP Transformation Cockpit] - Export Block Execution Monitor: BLK_SD_EXP_01              [Client 100]  |
+=============================================================================================================+
| Table | Reader | Total Records | Packages (Done/Total) | Processed MB | Status        | Speed (rec/s)  |
|-------+--------+---------------+-----------------------+--------------+---------------+----------------|
| VBAK  | SR     |       500,000 | 10 / 10  [COMPLETED]  | 142 MB       | [COMPLETED]   | 45,000         |
| VBAP  | SR     |     3,800,000 | 76 / 76  [COMPLETED]  | 890 MB       | [COMPLETED]   | 62,000         |
| KONV  | DR     |     8,500,000 | 170 / 170 [COMPLETED] | 1,820 MB     | [COMPLETED]   | 85,000         |
+=============================================================================================================+
```

# 03 Mapping Matrix & Number Range Strategies

Value conversions, composite rules, and target collision prevention

## 3.1 Transformation Mapping Framework

| Mapping Type | Scope | Example | Description |
|---|---|---|---|
| **1:1 Value Mapping** | Domain / Field | BUKRS 1000 -> 2000 | Direct one-to-one translation. |
| **N:1 Consolidation** | Domain / Field | BUKRS 1000, 1100 -> 2000 | Harmonizes multiple source codes into one target code. |
| **X+N:1 Dependent** | Composite Keys | WERKS 1000 + LGORT 0001 -> FG01 | Two source fields determine one target value. |
| **Derivation Rule** | Table Field | Profit Center derived from Plant | Dynamic ABAP logic calculating target value. |
| **Hard-Coded** | Table Field | VKORG = 'US01' | Static constant assigned unconditionally. |
| **Default Value** | Table Field | VSTEL = '1000' (if initial) | Assigned only when source field is initial/blank. |

**Validation Rule:** Always execute `/SNP/TE02_VALIDATE_MAPPINGS` before starting the import. Any unmapped value flagged in red must be resolved to prevent import failures.

## 3.2 Number Range Strategies

| Strategy | Mechanism | Best For |
|---|---|---|
| **Increment (API)** | Generates new numbers via NUMBER_GET_NEXT; logs cross-reference. | Master data (Customers, Vendors, Materials). |
| **Difference (+Offset)** | Calculates New = Old + Offset (e.g. +5,000,000). Fast, lock-free. | High-volume transactional documents (SD, FI, MM). |
| **1:1 Adoption** | Adopts source document number unchanged. | Dedicated target migration clients. |
| **Inactive** | Bypasses number range engine. | External numbering scenarios. |

# 04 Import Block, Modes & Lifecycle Events

Target insertion modes, field rule overrides, and commit sequencing

## 4.1 Complete Import Modes Matrix

| Import Mode | If Key Exists in Target DB | If Key Does NOT Exist | Typical Application |
|---|---|---|---|
| **INSERT (Default)** | Fails / Terminates (Duplicate Key) | Inserts new record | Strict cutover & clean target clients. |
| **INSERT_ADK** | Skips record silently (no error) | Inserts new record | Loading archive/historical master data. |
| **UPDATE** | Updates non-key fields | Skips record | Refreshing attributes on pre-existing data. |
| **MODIFY** | Updates non-key fields | Inserts new record | Delta synchronization & master data merge. |
| **DELETE** | Deletes target record | No action | Test data cleanup / Rollback blocks. |
| **APPEND / SUM** | Adds numeric amounts to existing row | Inserts new record | Trial balances / GL financial ledger merge. |

## 4.2 Import Events & Field Rules

Import rules run on the Target S/4HANA system. A **Field Rule** contains custom ABAP for a specific field and **overrides** any Global Import Rule for that field. Events allow dynamic mode switching per record (e.g., setting `TE_IMPORT_MODE = 'UPDATE'`).

### Transformation Cockpit Snapshot: Import Block & Mode Grid

```
+=============================================================================================================+
| [SNP Transformation Cockpit] - Import Block Configuration: BLK_SD_IMP_01                  [Client 200]  |
+=============================================================================================================+
| Seq  | Table Name | Target Table Name | Import Mode | Option | Target Redirect? | Status                  |
|------+------------+-------------------+-------------+--------+------------------+-------------------------|
| 0010 | ADRC       | ADRC              | INSERT      | NONE   | [ ] No           | [READY]                 |
| 0020 | KNA1       | BUT000            | MODIFY      | NONE   | [ ] No           | [READY]                 |
| 0030 | VBAK       | VBAK              | INSERT      | NONE   | [ ] No           | [READY]                 |
| 0040 | VBAP       | VBAP              | INSERT      | NONE   | [ ] No           | [READY]                 |
| 0050 | KONV       | PRCD_ELEMENTS     | MODIFY      | NONE   | [X] Redirect     | [READY]                 |
+=============================================================================================================+
```

# 05 Reconciliation, Consistency & Transaction Matrix

Post-import verification, FK checks, and master transaction reference

## 5.1 Post-Import Verification & Foreign Key Integrity

- **Record Count Reconciliation:** Validate that Source Exported Count = Target Inserted Count + Skipped (EV_SKIP) + Filtered Rows.
- **Foreign Key Integrity Check:** Execute transaction `/SNP/TE02_FK_CHECK` on target to ensure zero orphan records exist.
- **Selective Restartability:** If a package fails, fixing the root cause allows restarting *only failed packages* without duplicate inserts.

### Transformation Cockpit Snapshot: Reconciliation & Foreign Key Check

```
+=============================================================================================================+
| [SNP Transformation Cockpit] - Post-Import Reconciliation & Consistency Check            [Client 200]  |
+=============================================================================================================+
| Table | Source Exported | Target Inserted | Skipped (EV_SKIP) | Discrepancy | Status                    |
|-------+-----------------+-----------------+-------------------+-------------+---------------------------|
| VBAK  |         510,000 |         500,000 |            10,000 |           0 | [BALANCED - MATCH]        |
| VBAP  |       3,850,000 |       3,800,000 |            50,000 |           0 | [BALANCED - MATCH]        |
| KONV  |       8,500,000 |       8,500,000 |                 0 |           0 | [BALANCED - MATCH]        |
+-------------------------------------------------------------------------------------------------------------+
| Foreign Key Check (/SNP/TE02_FK_CHECK): Parent VBAK <-> Child VBAP | Total Orphans: 0 | Status: [PASSED]|
+=============================================================================================================+
```

## 5.2 Key Transaction Codes & Programs Matrix

| Transaction / Program | Technical Name | Core Purpose |
|---|---|---|
| **Transformation Cockpit** | /SNP/TC or /SNP/TE02 | Main UI for Data Objects, Selections, Rules, Mappings, Blocks. |
| **Daemon Controller** | /SNP/TE01_DAEMON | Starts background compiler and RFC worklist coordinators. |
| **Validate Mappings** | /SNP/TE02_VALIDATE_MAPPINGS | Pre-import validation scanning for unmapped source values. |
| **Automated Block Runner** | /SNP/TE02_API_EXECUTE_BLOCKS | API/CLI engine for scheduled background block runs. |
| **Table Catalog to CSV** | /SNP/TE02_TABLES_TO_CSV | Exports complete table catalogs and rules to CSV. |
| **Data Object Builder** | /SNP/TF_CO_DOB_700_DIA_ALL | Creates and configures Data Objects for custom Z* tables. |
| **Foreign Key Check** | /SNP/TE02_FK_CHECK | Validates referential integrity across imported target tables. |

**End of Document.** Master technical reference generated for SNP CrystalBridge Migration Architecture.